THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 11: **Electricity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): A voltmeter is connected in parallel with a resistor. Reason (R): A voltmeter has a very high resistance. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. The resistance of a wire of length L and area A is 12 ohm. A second wire of the same material has length 2L and radius double that of the first. Its resistance is 1

(a) 24 ohm **(b)** 12 ohm **(c)** 6 ohm **(d)** 3 ohm

3. Which graph is a straight line through the origin for a metallic conductor at constant temperature? 1

(a) V against t **(b)** V against I **(c)** I against R **(d)** P against I

4. Assertion (A): Copper wires are used for transmission of electricity. Reason (R): Copper has high resistivity. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A wire is stretched to three times its length with its volume unchanged. The new resistance is 1

(a) R / 9 **(b)** 9 R **(c)** 3 R **(d)** R / 3

6. In the circuit, ammeter reads 2 A and voltmeter across the resistor reads 10 V. The resistance is 1

(a) 20 ohm **(b)** 12 ohm **(c)** 0.2 ohm **(d)** 5 ohm

7. Assertion (A): The resistance of a conductor increases with temperature. Reason (R): At a high temperature, the electrons collide more often with the atoms. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A cylindrical wire of resistance R is cut into n equal parts which are joined in parallel. The equivalent resistance is 1

(a) n R **(b)** R / n^2 **(c)** R / n **(d)** n^2 R

9. **Assertion (A):** An ammeter is connected in series in a circuit. 1
Reason (R): An ammeter has a very low resistance.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The resistance of a conductor is directly proportional to its length. 1
Reason (R): The resistivity of a material depends on its temperature.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Calculate the number of electrons flowing per second in a wire carrying 4.8 mA. 2

12. A heater of 800 W is connected to a 220 V source. Find the resistance and the heat produced in 5 minutes. 2

13. How does the resistance of a 100 W bulb compare with a 25 W bulb of the same voltage rating? 2

14. A wire of resistance 4 ohm is bent to form a circle. Find its resistance between two diametrically opposite points. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. The resistance of a copper wire is 2.5 ohm at 20 degrees Celsius. Why does its resistance rise at high temperature, and what is the effect on the current if the voltage is constant? 3

16. Three bulbs of 100 W each rated 220 V are connected in parallel with a 220 V source. Find the total current and the equivalent resistance. 3

17. Why are ammeters of low resistance and voltmeters of high resistance? 3

OR

Two wires of the same material and length have radii in the ratio 1 : 2. Find the ratio of their resistances and of the currents when they are connected in parallel across a battery. 3

18. A student connects a voltmeter in series with the resistor by mistake. What will happen and why? 3

OR

For a given resistor, the current is 0.2 A for 4 V. If the voltage is increased to 10 V, find the new current and power. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In an experiment, a student connects a nichrome wire to a variable battery and notes the readings: when the potential difference is 2 V the current is 0.4 A; at 4 V it is 0.8 A and at 6 V it is 1.2 A. 5
(i) Find the resistance of the wire.
(ii) What is the shape of the V-I graph?
(iii) State Ohm's law.
(iv) The wire is cut into two equal halves and the two halves are joined in parallel. Find the resistance of the combination.

20. A wire of resistance 10 ohm is stretched uniformly to three times its length. Find the new resistance. The stretched wire is then folded into three equal parts and joined in parallel. Find the final resistance. 5

OR

Two bulbs rated 60 W and 40 W at 220 V are connected in series and the supply is 220 V. Find the current, the voltage across each and the power consumed by each. 5

END OF QUESTION PAPER Total: 40 Marks